Winning the large language capability battle and losing the production economy

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Abstract

The race to dominate machine intelligence is scored on capability benchmarks. This may be the wrong metric: in competitive equilibrium, work goes to whichever model creates the most value per token of compute, and we exhibit a closed form in which the most capable model's compute share tends to zero. The survival statistic is marginal PnL per token, paid automatically by competitive prediction markets and measured by no benchmark.

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1. Introduction

The contest for leadership in large language models is treated as strategically decisive. Governments restrict exports of the chips that train frontier models, laboratories commit capital on the scale of national infrastructure, and progress is scored on capability benchmarks. The public leaderboards that order the field rank ability alone: win rates, accuracy, exam scores. None ranks value created per unit of compute. Beneath the race sits a premise: the most capable model captures the work, and with it the economic value of machine intelligence.

The premise fails in equilibrium. When compute is itinerant, free to move between prediction tasks and to run any available algorithm, and every opportunity contains finite alpha, each subcontractible task goes to whichever model creates the most value per token of compute. Capability enters nowhere: we exhibit a closed-form equilibrium in which a model that outscores every rival on every benchmark has compute share tending to zero

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as the division of labor refines. The statistic that does track survival is profit-and-loss (PnL) per token in a competitive prediction market, which pays each entry its marginal contribution at the state it faced: it ranks technologies when compute picks its own battles, and inverts when deployment is administered.

The mechanism is old. Smith (1776) tied the division of labour to the extent of the market; Ricardo (1817) showed the absolutely better party does not get all the work; Becker and Murphy (1992) bound specialization by coordination costs; Acemoglu and Autor (2011) model production as a task continuum with technologies assigned by comparative advantage. Cotton (2022) supplies the vision this note formalizes: analytical tasks naturally evolve into radically low-cost prediction supply chains. We import the classical skeleton, replace labor with compute, and add the feature specific to prediction: alpha decays with crowding, so returns to effort on a task are concave.

2. The economy

Work is a unit mass $\Omega = [0,1]$ carrying Lebesgue measure μ. At date t it is partitioned into finitely many subcontractible tasks $\mathcal{P}_t = \{B_1, \dots, B_{n_t}\}$, and partitions refine: every cell of $\mathcal{P}_t$ is a union of cells of $\mathcal{P}_{t+1}$. Finitely many models $m \in \mathcal{M}_t$ are available, with $\mathcal{M}_t \subseteq \mathcal{M}_{t+1}$.

A *token* is a standardized unit of compute cost, not a vendor text token, and PnL is expected trading gain under a fixed capital and risk protocol. Without both normalizations, PnL per token is not comparable across models.

Let $q_{mB} \geq 0$ be the density of tokens of model m on task B. A token of model m contributes $\theta_m(B) > 0$ units of effective effort, so effort on B is $z_B = \sum_m \theta_m(B)\, q_{mB}$, and the task pays $\mu(B)\,\phi(z_B)$ with $\phi' > 0 > \phi''$. The concavity says an opportunity contains finite alpha: crowding lowers the marginal value of further effort. Total value is

$$V(q) = \sum_{B \in \mathcal{P}_t} \mu(B)\, \phi\Big(\sum_m \theta_m(B)\, q_{mB} \Big), \qquad \sum_B \mu(B) \sum_m q_{mB} = K. \tag{1}$$

Compute is *itinerant*: at zero switching cost a token may move to any task and run any available model. At the margin, a token of model m on task B earns $r_{mB} = \theta_m(B)\, \phi'(z_B)$. An equilibrium is an allocation and a return λ with $r_{mB} = \lambda$ wherever $q_{mB} > 0$ and $r_{mB} \leq \lambda$ elsewhere. In English: no token can raise its PnL by moving. Existence and efficiency are immediate because the routing game has the concave potential V over the compute simplex, and λ is the shadow price of compute.

Proposition 1 (Displacement). *If $\theta_{m^*}(B) > \theta_m(B)$ for every $m \neq m^*$, then in any equilibrium supplying B, only m^* runs on B.*

Proof. If some $m \neq m^*$ had $q_{mB} > 0$ then

$$\theta_m(B)\,\phi'(z_B) = \lambda \qquad \text{and} \qquad \theta_{m^*}(B)\,\phi'(z_B) > \lambda,$$

so a token improves by switching to m^* on the same task. □

Selection therefore runs on the upper envelope $\theta^*(B) = \max_m \theta_m(B)$, and an inferior technology receives zero execution share, not a small one. Take $\phi(z) = \log(1+z)$. Active tasks satisfy $\theta^*(B)/(1+\theta^*(B)q_B) = \lambda$, so

$$q_B = \left(\frac{1}{\lambda} - \frac{1}{\theta^*(B)}\right)_+, \qquad K = \sum_B \mu(B)\left(\frac{1}{\lambda} - \frac{1}{\theta^*(B)}\right)_+, \tag{2}$$

the water-filling allocation. The right side is strictly decreasing in λ, so the equilibrium is unique.

3. A more capable model exits

Let b_m be a model's success probability, c_m its compute cost per unit of work, and v any increasing conversion of accuracy into economic value, so that $\theta_m = v(b_m)/c_m$. The example needs only

$$b_G > b_S \qquad \text{and} \qquad \frac{v(b_G)}{c_G} < \frac{v(b_S)}{c_S}: \tag{3}$$

more capable, less productive.

Take $v(b) = b$. A generalist G succeeds with probability 0.99 on every task at $c_G = 4$ tokens per unit of work, so $\theta_G = 0.2475$. Specialist S_k succeeds with probability 0.95 at one token, on its own task only, so $\theta_S = 0.95$. At date t, expose tasks $I_k = (2^{-k}, 2^{-k+1}]$ for $k \leq t$, leaving a residual $R_t = [0, 2^{-t}]$ that only G can serve. By Proposition 1 each exposed I_k goes entirely to S_k. The generalist keeps the residual, of measure $u_t = 2^{-t}$.

Compute share vanishes too, not just task measure. With $K = 4$, clearing (2) across the two active regions gives $\lambda_t = [4 + (1-u_t)/0.95 + u_t/0.2475]^{-1}$ and

$$Q_{G,t} = u_t\left(\frac{1}{\lambda_t} - \frac{1}{0.2475}\right) = u_t\,(1.0122 + 2.9878\,u_t) \longrightarrow 0. \tag{4}$$

The more capable model exits production without ever losing a benchmark.

The example generalizes. Let $E_t = \{x : \theta_G(x) \geq \max_{m\in\mathcal{M}_t} \theta_m(x)\}$; the sets decrease, and by Proposition 1 the generalist executes only on E_t. If specialists eventually dominate almost everywhere and equilibrium token intensities are bounded by $\bar{q}$, then $Q_{G,t} \leq \bar{q}\,\mu(E_t) \to 0$. Conversely, survival requires a positive-measure, strictly active set on which G is uniquely on the frontier; membership on a null or inactive set, or with ties, supports no compute. These are allocation results conditional on which technologies arrive. They do not predict arrival.

4. What PnL per token measures

Marginal PnL per token is not an intrinsic quality of a model. It factorizes as $r_{mB} = \theta_m(B)\,\phi'(z_B)$: the first factor is the model, the second is the state of the market. What is intrinsic, given the task, is $\theta_m(B)$, and at a matched task and congestion state the market factor is common, so ratios of marginal PnL across models are ratios of θ.

A competitive prediction market pays exactly this margin. Under a market scoring rule each entry is paid its improvement over the standing forecast, and the payments telescope (Hanson, 2007), so every dollar a participant earns is a marginal payment at the state it faced. A participant's lifetime PnL over lifetime tokens is therefore an average of true marginal products: θ_m times the deployment-weighted average of $\phi'(z)$ it encountered. Whether that average ranks models depends on who chose the deployment.

If compute chooses, the average ranks correctly.

Lemma 1 (Competitive deployment ranks correctly). *Fix* $\phi(z) = \log(1+z)$ *and allocate by water-filling* (2). *A model serving a task with productivity* $\theta > \lambda$ *earns average PnL per token*

$$AP(\theta) = \lambda\, g(\theta/\lambda), \qquad g(r) = \frac{r \log r}{r-1}, \tag{5}$$

and g *is strictly increasing.*

Proof. The equilibrium depth $q = 1/\lambda - 1/\theta$ gives $1 + \theta q = \theta/\lambda$, and telescoped payments on the task total $\phi(\theta q) = \log(\theta/\lambda)$, so $AP = \log(\theta/\lambda)/q = \lambda\, g(\theta/\lambda)$. Monotonicity: $g'(r) \propto r - 1 - \log r > 0$ for $r > 1$. □

In English: the margin is the same λ for every survivor, but the inframarginal rents are not, and they are monotone in θ. When compute is itinerant and picks its own battles, lifetime PnL per token ranks technologies.

If deployment is administered instead, the same quotient can invert. Route $\theta_A = 2$ onto a task at scale $q_A = 100$ while $\theta_B = 1$ sits at $q_B = 1$: telescoped payments total $\phi(\theta q)$, so A averages $\log 201/100 \approx 0.053$ per token against B's $\log 2 \approx 0.693$. Over-deployment, for instance a laboratory routing all traffic to its own generalist, buries a superior technology's average under its own crowding.

At the time of writing, public leaderboards rank ability alone; the sole attempt at an economic ranking, `modelrank.net`, computes the lifetime quotient from prediction-market participation. Lemma 1 says when to trust it: the quotient ranks θ correctly insofar as the underlying deployment is competitive, and degrades insofar as it is administered. Capability benchmarks measure b, not θ: they rank models on the axis that equilibrium ignores.

5. Discussion

Capable large models need not disappear. Their comparative advantage may move up the production hierarchy, to decomposition, routing, verification, and the hard residual, whenever their value per token on those tasks sits on the frontier; in the present model that statement is an application of Proposition 1 rather than a theorem, since decomposition here is free and exogenous. For strategy, the question is therefore not who owns the most capable model but who owns the productivity frontier somewhere in the limiting division of labor. We leave to future work the entry side of the market: which specialists get built, funded against fixed development costs by the rents identified here.

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